

Individual Assignment 5

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1. Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

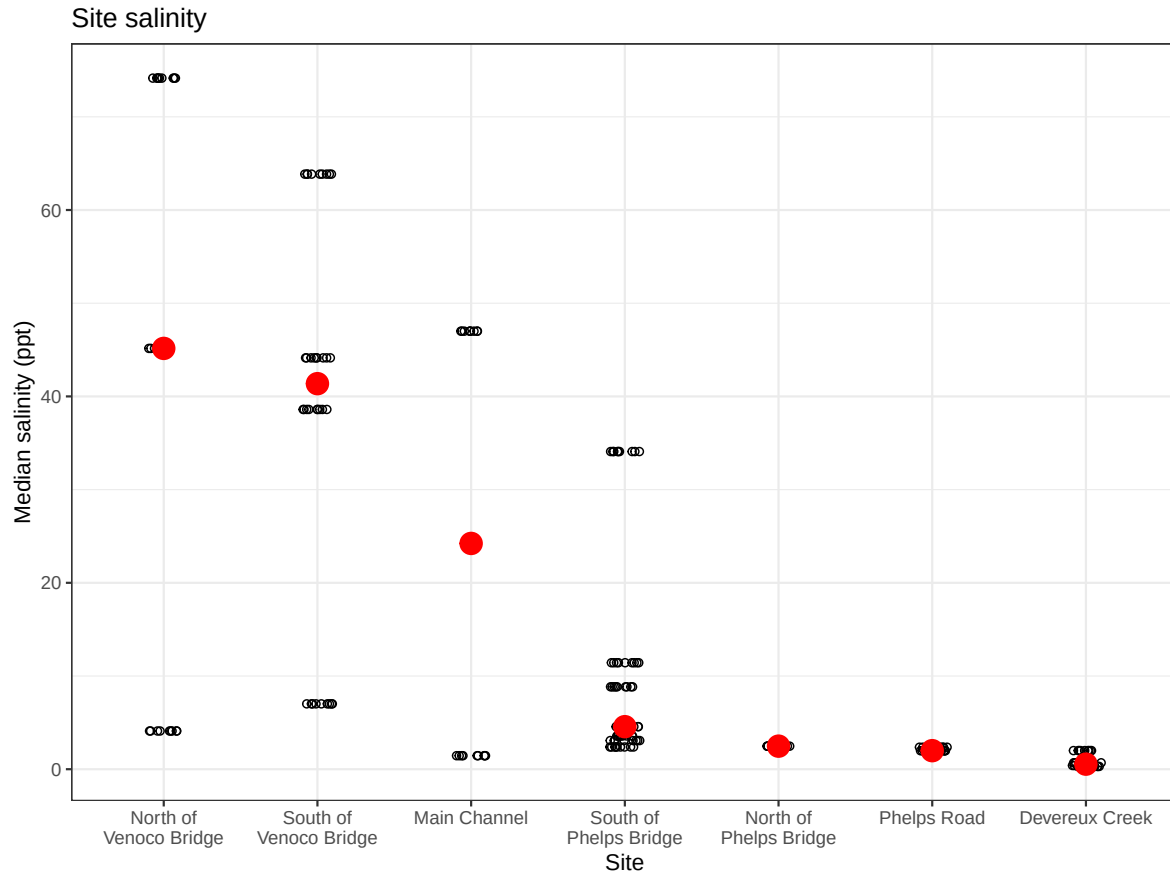
Importing other graphs

```

# base layer
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
  color = "red",
  size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
  y = "Median salinity (ppt)",
  title = "Site salinity") +
# different theme
theme_bw()

# displaying object
salinity

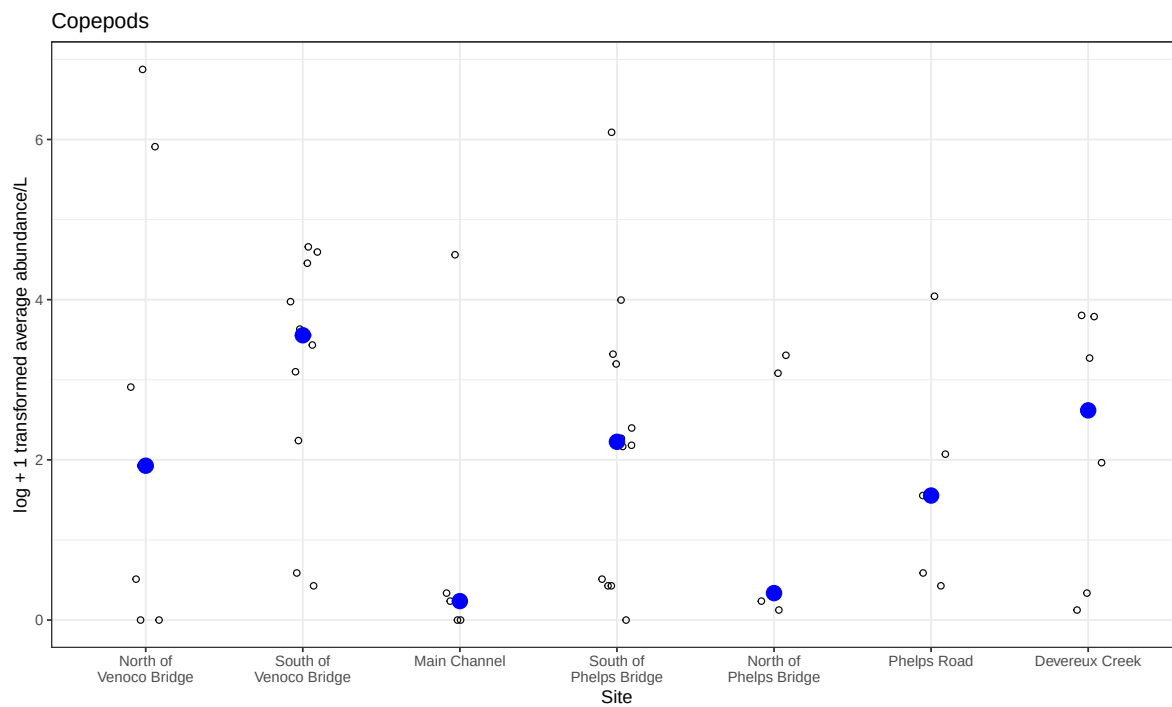
```



```
# base layer
copepods_plot <- ggplot(data = copepods_df,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
    fun = "median",
    size = 4,
    color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
```

```
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Copepods") +
# different theme
theme_bw()

# displaying object
copepods_plot
```



Question 1

A: Planning

- x axis: Site
- y axis: log transformed Ostracod abundance
- geometry abundance: `geom_jitter()`
- geometry median: `stat_summary()`

B: Wrangle Data

```
#new object from aq_ins_clean
ostracod <- aq_ins_clean |>
  # filter to only include ostracod
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows
slice_sample(ostracod, n=10)
```

```
# A tibble: 10 x 24
  date_site      date            site taxon ave_lit med_ph med_sal med_do
  <chr>          <dtm>          <chr> <chr>   <dbl>   <dbl>   <dbl>   <dbl>
1 2020-01-21_NDC 2020-01-21 00:00:00 NDC    ostr~ 2.4 e+0    6.53    0.3    7.6
2 2022-08-12_NEC 2022-08-12 00:00:00 NEC    ostr~ 0          7.37    NA     NA
3 2022-05-06_NMC 2022-05-06 00:00:00 NMC    ostr~ 0          9.96    1.45   16.5
4 2022-05-12_NPB2 2022-05-12 00:00:00 NPB2   ostr~ 1.20e+3   NA     NA     NA
5 2023-09-15_NMC 2023-09-15 00:00:00 NMC    ostr~ 4 e-1     NA     NA    14.2
6 2022-02-23_NPB1 2022-02-23 00:00:00 NPB1   ostr~ 0          NA     NA     NA
7 2023-03-03_NDC 2023-03-03 00:00:00 NDC    ostr~ 5.73e+0    7.45    0.688   5.64
8 2022-05-05_NEC 2022-05-05 00:00:00 NEC    ostr~ 2.67e-1    9.45   63.8    7.3
9 2022-01-28_NDC 2022-01-28 00:00:00 NDC    ostr~ 9.47e+0    7.65    2.00   9.55
10 2023-07-17_NPB 2023-07-17 00:00:00 NPB    ostr~ 4.53e+0   NA     NA     NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

C: Code Figure

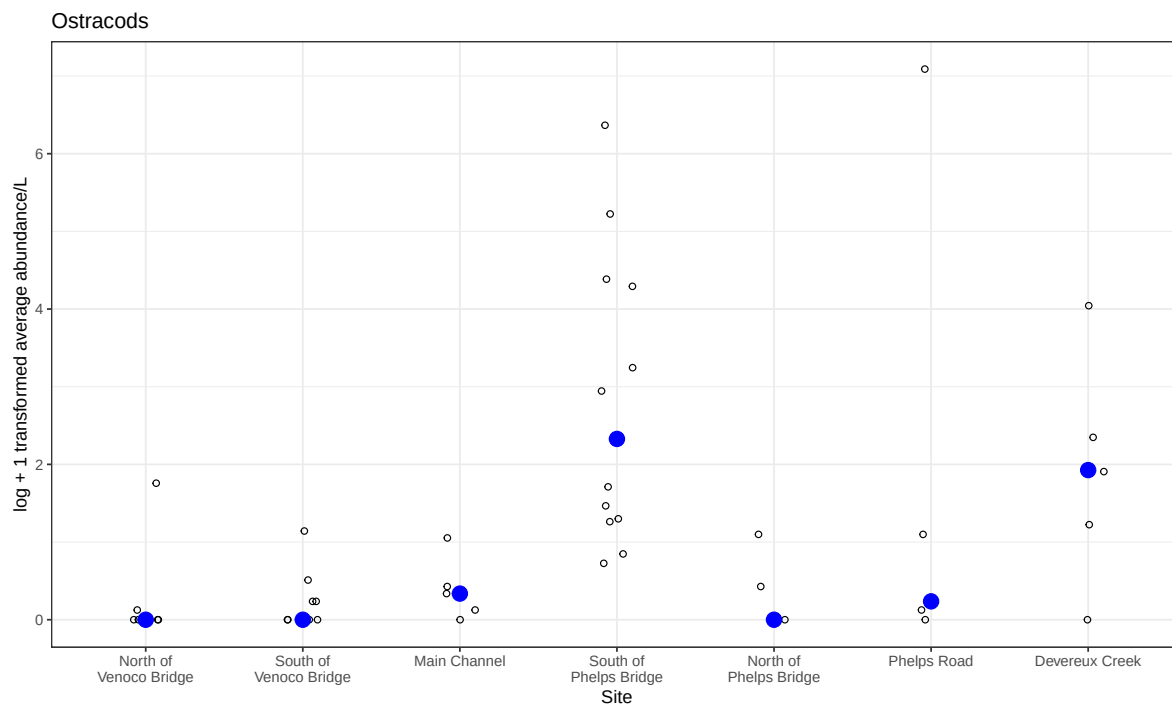
```
# base layer
ostracods_plot <- ggplot(data = ostracod,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent observations
```

```

geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +
# different theme
theme_bw()

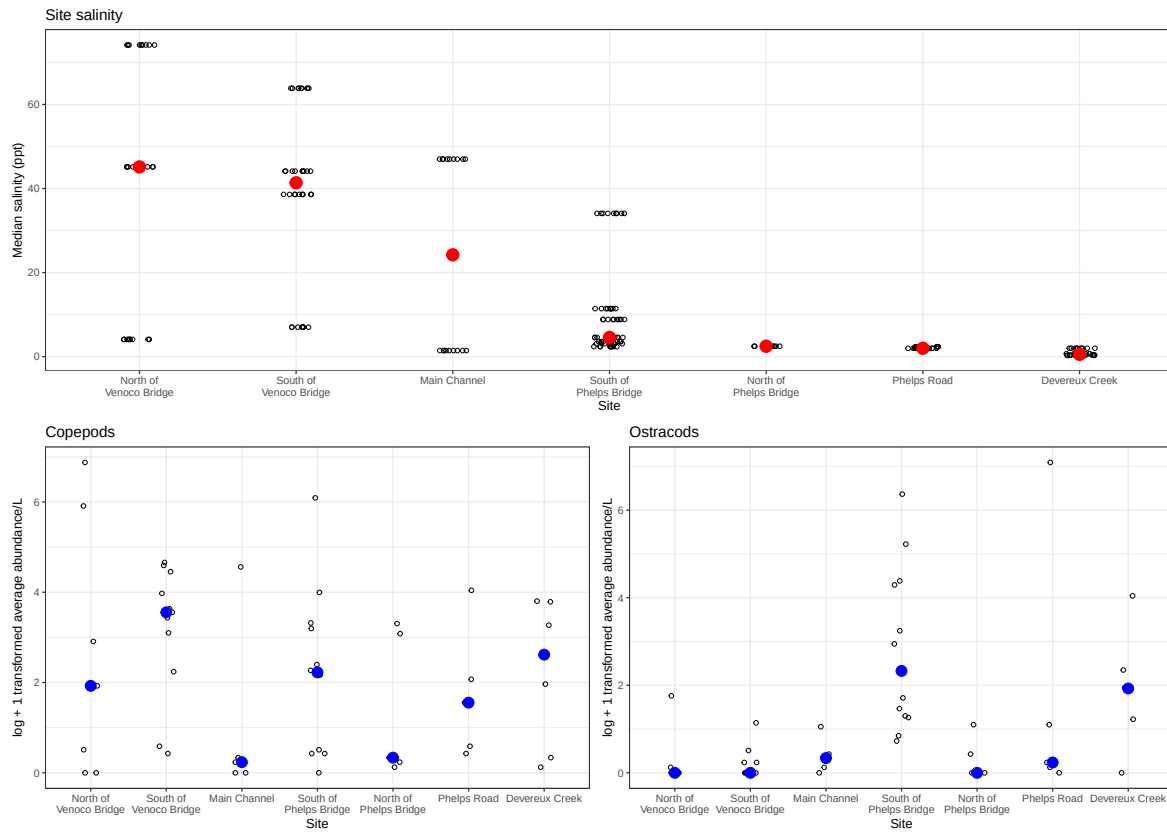
# displaying object
ostracods_plot

```



D: create multipanel figure

```
#combining plots  
salinity/(copepods_plot+ostracods_plot)
```



E: Analysis

Question 2

A: Planning

- x axis: salinity
- y axis: log transformed abundance
- color: taxon
- geometry: `geom_point()`
- facets: `facet_Wrap(~taxon, scales="free")`

B: Wrangle Data

```
# creating new clean object from aquatic inverts
ab_salinity <- aq_ins_clean |>
  # log transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit+1),
    # convert taxon to title
    taxon = to_any_case(taxon, case = "title"),
    # order taxon levels by abundance
    taxon = fct_reorder(taxon, ave_lit, .fun="max"))

#display 10 random rows
slice_sample(ab_salinity, n = 10)
```

A tibble: 10 x 24

	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>
1	2022-01-24_NEC	2022-01-24 00:00:00	NEC	Cope~	1.05e+2	NA	NA	NA
2	2022-01-21_NEC	2022-01-21 00:00:00	NEC	Cope~	9.8 e+1	NA	NA	NA
3	2023-11-11_NEC	2023-11-11 00:00:00	NEC	Nema~	0	NA	38.6	0.25
4	2023-09-15_NMC	2023-09-15 00:00:00	NMC	Hemi~	2.67e-1	NA	NA	14.2
5	2022-05-12_NPB2	2022-05-12 00:00:00	NPB2	Ostr~	1.20e+3	NA	NA	NA
6	2023-08-12_NEC	2023-08-12 00:00:00	NEC	Anne~	0	NA	NA	NA
7	2023-02-12_NEC	2023-02-12 00:00:00	NEC	Hemi~	2 e+0	NA	NA	NA
8	2024-01-28_NPB	2024-01-28 00:00:00	NPB	Ostr~	3.33e+0	7.43	3.08	6.6
9	2022-02-21_NPB	2022-02-21 00:00:00	NPB	Clad~	3.2 e+0	7.39	34.1	0.34
10	2022-01-28_NDC	2022-01-28 00:00:00	NDC	Dipt~	0	7.65	2.00	9.55

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```